Collaborative FMCW Radar Swarm with Onboard AI Decision Engine for Aerial Surveillance

Document 4 -- Additional Infor

Aran Technologies | aranrobotics@gmail.com | IAF MBC-3 Competition

1. Phase I Testing -- Completed

Five-drone swarm simulation verified on ROS2 Jazzy / Gazebo Harmonic: ASP generation, bully-protocol leader election, graceful degradation, sector reassignment -- 344 tracks logged, four surviving drones sustaining full continuity. Pre-recorded 4-min ISR video (1920x1080, v12-MPC-v5) ready: 11-WP survey, 50 m orbit +/-0.5 m, RTL, 3D map save. Broadband and drone-to-ground comms tested and operational.

2. Indigenisation Breakdown

Sub-system	Indigenous Content
Hexacopter airframe	Indigenous design and local fabrication
Antenna panels (6 x per drone)	Indigenous PCB design
AI software stack (CFAR, RF, LLM)	100% indigenous (Python / C++, open-source models)
GCS dashboard (Flask / SocketIO)	100% indigenous
ROS2 swarm coordination stack	100% indigenous
Flight controller firmware	Custom STM32 PCB -- designed from scratch
Compute module (Jetson)	COTS -- imported
Radar frontend (AWR1843)	COTS -- imported
Mesh radio (Doodle Labs)	COTS -- imported

Estimated indigenisation: >= 55% by mission-criticality weighting -- meets MBC-3 sec.2.25 requirement.

3. Team

Aran Technologies is a three-member engineering startup building indigenous defence UAS systems from the ground up.

Kishore Udhayakumar | Founder & CEO -- Mechanical Engineering. Product development, mechanical design, thermal engineering, business strategy, and defence market engagement.

Vegi Tejo Bhargav | Co-Founder & CTO -- Mechanical Engineering. Project execution, hexacopter airframe fabrication, additive manufacturing, and hardware integration.

L. Harsha Vardhan Naidu | Product Head, Electronics -- EEE. Autonomous mission stack: PX4 SITL, ROS2 swarm middleware, GCS dashboard, STM32 FC firmware, and three-layer AI pipeline (CFAR -> RF -> LLM).

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